SHINING 3D

# RobotScan Control

V2.3.0

# User Manual

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# Overview

## Welcome

This user manual (hereinafter referred to as "the Manual") introduces the operation of RobotScan Control.

### Symbol Conventions

| Symbol | Meaning |
|---|---|
| | **Note**: This symbol is used to inform you of the additional information of the product. |
| | **Caution**: This symbol is used to inform you of incorrect operations that may damage the device or result in data loss. Any damages resulting from misuse are not covered by the warranty. |
| | **Warning**: This symbol is used to inform you of the potential risks that may result in serious personal injury and other safety incidents. |

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# Quick Guide

This chapter provides an overview of the RobotScan Control software, helping you quickly find the relevant operational assistance content.

Here you can learn about the software's functions, installation, and other related content.

→ What kind of software is this?

→ How to install the software and what are the environmental requirements?

→ General introduction to the software interface

**Once you have successfully installed the software, the complete process for using the software is as follows:**

**Add Device**

**Arrange Workflow**

Before designing the automation workflow, you need to add and connect the device.

→ How to add and connect the device?

After connecting the device, design the workflow and execute the automation tasks.

→ How to arrange the automation workflow?

# Software

## Introduction

RobotScan Control is an automated software platform that seamlessly integrates 3D scanners, collaborative robots, and hardware devices with PLC support. Its visual workflow design let users build automation processes without programming. It can continuously captures 3D data, connects with inspection software, and automatically generates standardized reports — simplifying and streamlining high-precision 3D scanning and batch inspection.

### Software Applications

### Aerospace

Inspect the shape and dimensions of high-precision components and complex surfaces to ensure product safety and reliability.

### Automotive Industry

Conduct regular 3D inspections during the production process to monitor the stability of the production line and the consistency of product quality.

### 3C Electronics

Convert actual parts into digital models through 3D scanning and measurement for product design modifications, fault analysis, and more.

### Precision Hardware

Perform full-size inspections and tolerance analysis on precision machined parts to ensure machining accuracy and product qualification rates.

### Precision Injection Molding

Achieve size inspection and deformation analysis of molds and injection parts, aiding in product quality improvement and process optimization.

# Software Installation

## Operating Environment

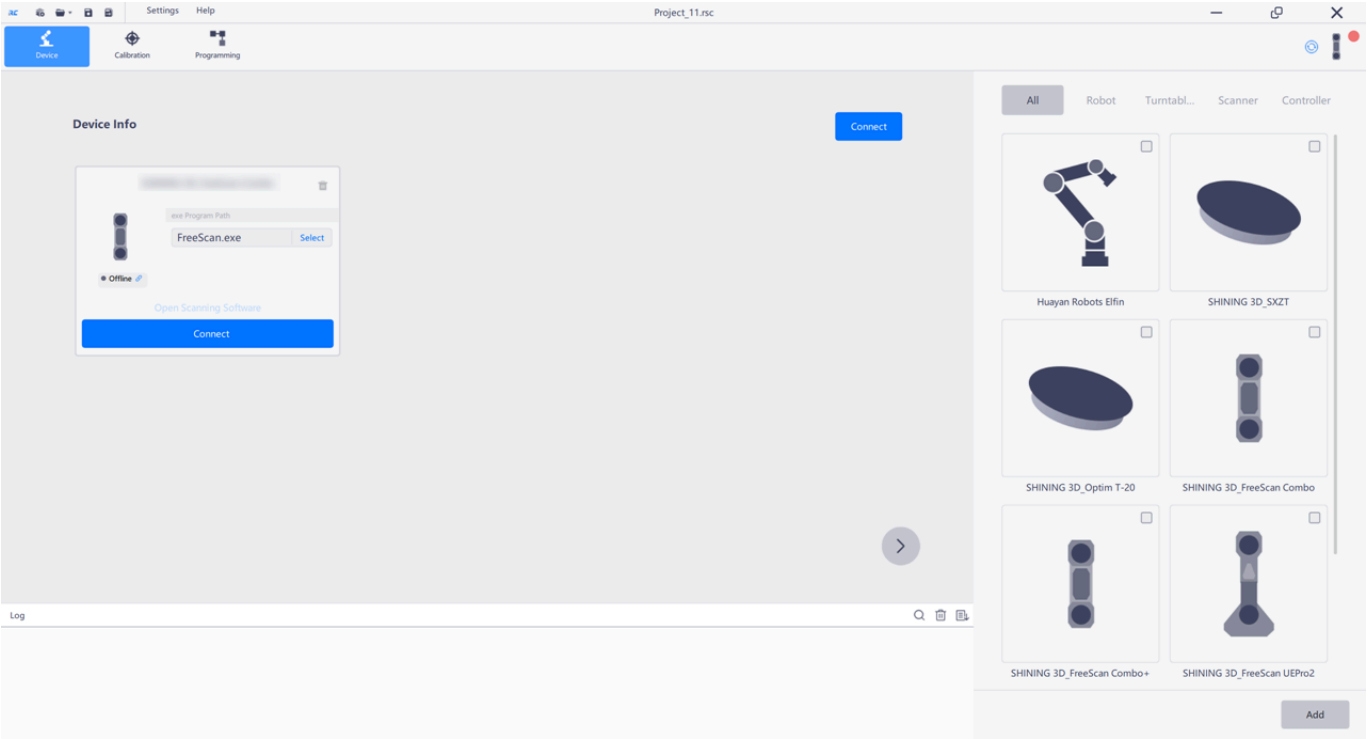
| Recommended Configuration | |
|---|---|
| Processor | Intel® Core™ i7-13700H or higher |
| Graphics Card | NVIDIA RTX 4060 Laptop GPU or higher |
| Video Memory | 8 GB or more |
| RAM | 64 GB or more |
| Operating System | Windows 11 (64-bit) |

**Note**

- To ensure the software runs properly, please use an NVIDIA graphics card.
- Administrator privileges are required for software installation. The initial installation may take some time, so please be patient.
- Do not install the software in the **C:\Program Files or C:\Program Files (x86)** directories to avoid issues with permissions that may prevent the software from starting properly.

# Software Interface

## Overview




### Menu Bar

#### ① File

Supports **New**, **Open**, **Save**, and **Save As Solution** (*.rsc).

To open recent files, click on to the right of , which will expand the recent files list.

**Note**

- The default name is **Project_1**, and each time a new solution is created, the number increments by one.
- The default path is **C:\Users\Current Logged-in User's Username\Documents\RobotScan Control\Solution**, which can be modified in **Settings**.

### ② Settings

| Setting Item | Description |
|---|---|
| Select Language | Change the interface language, supporting Chinese, English, and German. |
| Project Save Path | Modify the default path for the project. Folders requiring administrator permissions are not supported. |
| Automatically start at boot time | / |
| Starting the software opens the previous project by default | / |

### ③ Help

User Manual: Click to open this user manual.

Shortcut operations: For details, see Shortcut Operations.

### Log

Display execution information, formatted as `yyyy-MM-dd HH:mm:ss log level log message`.

## Device Configuration

Used to add, configure, and connect devices; after connecting the device, you can view the device connection status.

You can save the current solution as a template for quick application.

## Calibration

To calibrate the device's scanning accuracy and maintain the quality of the scans.

## Programming

To create and manage workflows, utilize the visual programming interface. Drag-and-drop nodes onto the canvas and connect them to design complex scanning and inspection routines.

# Shortcuts

| Global | |
|---|---|
| Save Project | ^ Ctrl + S |
| Copy | ^ Ctrl + C |
| Paste | ^ Ctrl + V |
| Cut | ^ Ctrl + X |
| Delete | ⌦ Del |
| Group | ^ Ctrl + G |
| Ungroup | ^ Ctrl + ⌫ Backspace |
| Rename Group | F2 |
| **Node View & Scan View** | |
| Zoom | Scroll Wheel |
| Pan | Middle Mouse Button |
| Vertical Scroll | ^ Ctrl + Scroll Wheel |
| Horizontal Scroll | ⇧ Shift + Scroll Wheel |
| Multi-select | ⇧ Shift + Left Click |
| Box Select | Left Click Drag |
| **Inspection Report** | |
| Zoom | ^ Ctrl + Scroll Wheel |

# Device Management

# Device Management

RobotScan Control supports the connection and control of various models of robots, scanners, turntables, and other devices.

## Add Device

1. Click the **Device** button in the upper left corner of the interface.
2. Select the desired device type on the right, such as **turntable**.

| Device Type | Allowed Quantity |
|---|---|
| Robot | 1 |
| Turntable | 1 |
| Scanner | 1 |
| PLC | 1 |

3. Check the device you want to add, then click the **Add** button.

## Connect Device

1. Configure device information.

- **Huayan Robot**: Enter the robot's IP address (the default port is 10003 and cannot be modified).

> **Note**
>
> - The computer's network port must be in the same subnet as the robot's IP address.
> - You can check the IP by logging into the robot's backend (default account: admin, default password: admin): Management > Network > Fixed Address.
> - After entering the IP address, click **Configure** to open the robot's backend management page (the entered IP address + dist, such as 192.168.40.10/dist).

- **Turntable**: Enter the zero offset (i.e., the offset value between the target angle and the real-time angle).

> **Caution**
>
> Modifying the zero offset will affect subsequent turntable programming.

- **Scanner**: Select the local path for the scanning software.

2. Click the **One-Click Connect** button, and after the prompt **Verification Successful**, the device will connect automatically. If the verification fails, please try again.

| Icon | Status |
|---|---|
| | Not Connected |
| | Connection Successful |

> **Note**
>
> - Before connecting the scanner, please ensure that the device has been authorized and activated.
> - If the scanner is offline, the software will automatically attempt to reconnect.

> **Caution**
>
> After the robot is successfully connected, it is not enabled. You need to click **Start**, and after waiting for the robot to power up completely, continue by clicking **Enable** to use it.

| Text | Status | Description |
|---|---|---|
| Open | Successfully connected to the robot, the robot is powered off | Click to power on the robot |
| Enable | Successfully connected to the robot, power on the robot successfully | After clicking, the robot can perform teaching and execute workflows |
| Enable (Disable) | Robot connected successfully, robot is enabled | After clicking, the robot cannot move |
| Close | Successfully connected to the robot, power on the robot successfully | Click to turn off the robot power |

# 3D Mouse

You can use the mouse to control the robot for quick translation and rotation operations during robot point recording, replacing manual teaching.

For more information, please refer to the 3Dconnexion User Manual .

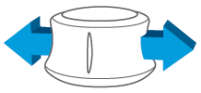
## Mouse Connection

Plug the USB receiver into the USB port of your computer, then you can use the SpaceMouse Wireless in the software.

## Operation Instructions

**Caution**

- The 3D mouse can only control the robot during point recording.
- Left-click to save the point position, right-click to exit recording.

| Illustration | Description |
| --- | --- |
|  | Control by pushing the button left and right:<br>• Robot end, X positive and negative translation<br>• Single-axis movement, first joint axis |
| | Control by pushing the button forward and backward:<br>• Robot end, Y positive and negative translation<br>• Single-axis movement, second joint axis |
| | Control by pushing the button up and down:<br>• Robot end, Z positive and negative translation<br>• Single-axis movement, third joint axis |
| | Control by tilting the button left or right:<br>• Robot end, RX positive and negative translation<br>• Single-axis movement, fourth joint axis |
| | Control by tilting the button forward or backward:<br>• Robot end Y-axis rotation, RY positive and negative translation<br>• Single-axis movement, fifth joint axis |
| | Control by rotating the button:<br>• Robot end X-axis rotation, RZ positive and negative translation<br>• Single-axis movement, sixth joint axis |

# Calibration

## Calibration Instructions

You can re-adjust parameters of the scanner through calibration, which ensures not only the accuracy of the device, but also the scanning quality.

**Note**

Calibration is required under the following conditions:

- The scanner was severely shaken or vibrated during transportation.
- The scanner is used for the first time or it is not in use for a long period of time (1-2 weeks).
- The accuracy drops during scanning, such as frequent alignment failures or the inability to recognize markers.
- Incomplete data is acquired or the quality of scanned data drops during scanning. severely decreased, such as frequent stitching errors or failure to recognize markers.

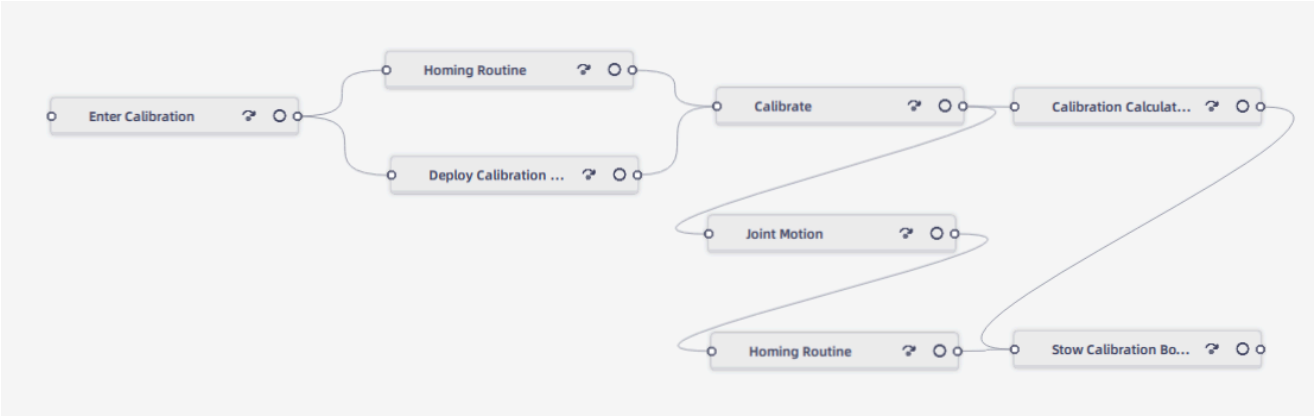
**Caution**

- Do not wipe the calibration board with chemical liquids.
- Do not put heavy objects or sundries on the calibration board.
- After using the calibration board, put it in the flannel bag.
- The calibration board can only be used for the calibration of the device.
- Keep the calibration board away from corrosives, metals and sharp objects to avoid corrosion or damage.
- Make sure that the markers on the calibration board aren't damaged or stained, and that the front side of the calibration board is clean and free of scratches.
- Make sure both the scanner and tracker are online before or during calibration.
- After powering on the device, please heat it to a proper temperature (about 35℃) before the calibration to ensure accuracy and precision.

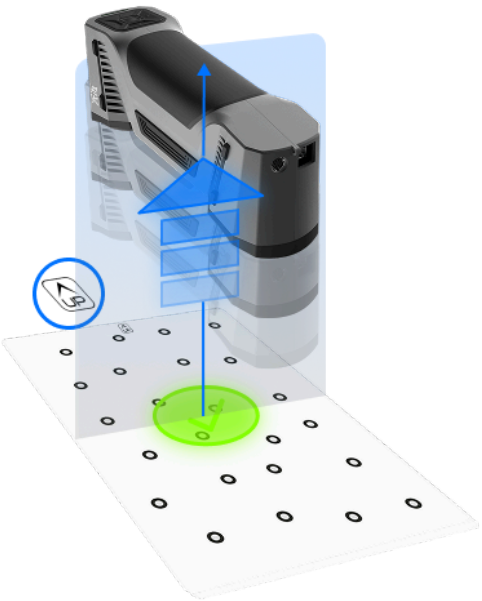
# Calibration Operation

Please follow the steps in the interface calibration guide.

1. Drag commands to orchestrate the calibrated workflow.




2. Click ▶ to start executing the command.
3. Place the calibration board into the base calibration plate when it pops out, with the front side facing up and laid flat.



The image is for illustration purposes only; the actual equipment may vary.

4. Switch to the **Handheld Calibration** tab to check the calibration progress.
5. During the calibration process, make sure to align the device's blue circle with the gray circle on the calibration board, and the calibration prompt box will turn green.
6. Calibration is complete once the prompt calibration is successful.

**Note**

- If the calibration is unsuccessful, you can modify the **joint movement** command's points and try again.
- If calibration still fails after multiple attempts, please contact the supplier or technical support.

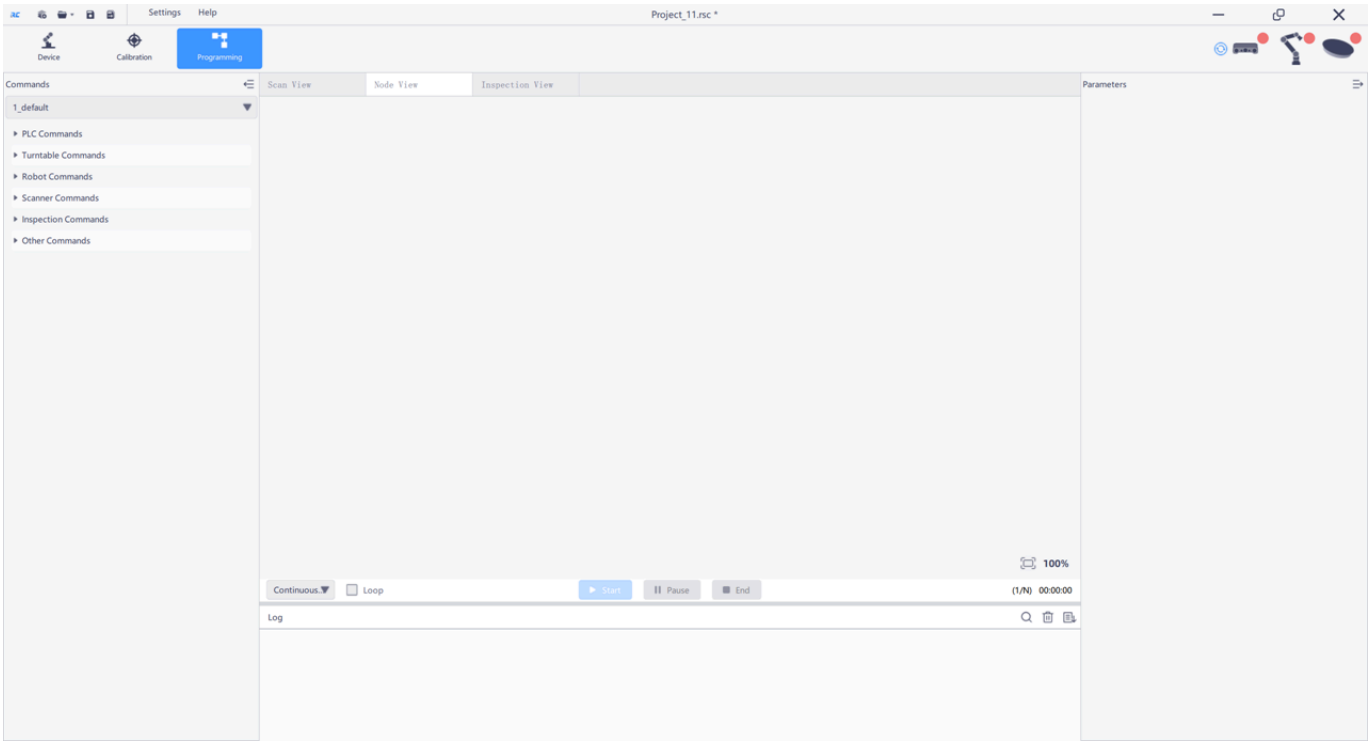
# Programming

Streamline your scanning and inspection tasks with Robotscan Control's powerful workflow engine. Visually design, connect, and edit commands with simple drag-and-drop actions.

# Example

Rapid Scan Template

# Interface



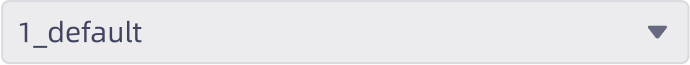

## Workspace

View: A view is a tab, and you can drag the view to become a separate window.

- Node view: Used to place and connect commands.
- Inspection View: Used to view the detection report.
- Scan view: Used to view scan results.

Command List: A list of all available commands from which you can drag commands onto the canvas.

Program List: Used to save and manage the commands you have assembled. You can view, create, or import different program files for easy reuse and sharing.



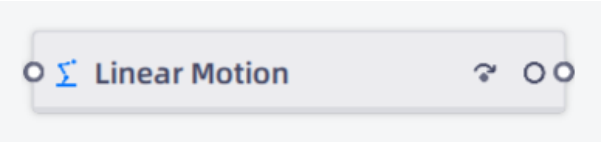

Parameter Panel: Displays the parameter settings of the selected command.

## Command Interface

All commands in Robotscan Control are based on a similar structure. This interface is suitable for any type of command.

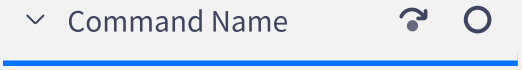
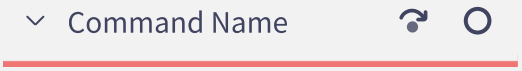
**Title**

The title displays the name/type of the command. On the right side of the title are the skip and breakpoint buttons for debugging.




**Status**

Below the title, the current status of the command is displayed, including not executed, executing, execution completed, and execution failed.

| Status | Example |
|---|---|
| Not Executed | |
| Executing |   |
| Execution Completed | |
| Execution Failed |   |

**Input**

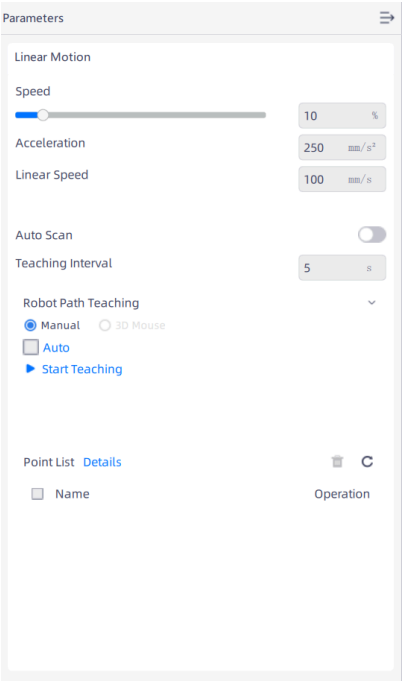
The input is on the left side of the command.

**Output**

The output is on the right side of the command and can be connected to the input of another command.

**Parameters**

Many command parameters may affect how they interact with input and output.




*Parameter for the robot's joint motion command.*

# Operations

## Create Command

Drag the command you need from the command list onto the canvas.

## Connect Commands

There are two ways to connect commands:

- Select the output port of one command on the canvas, drag it to the input port of another command, and release the mouse to complete the connection.
- Click on a port to create a connection line, move it to a blank area on the canvas, click again to bring up the command list, and select a command to complete the connection.

Some commands can accept multiple inputs or outputs; you can connect multiple commands to the same command or connect the output of one command to multiple commands.

## Edit Command

Click on a command on the canvas, or use ⇧ Shift + left-click / drag to select multiple commands, then right-click to choose an action from the menu: step execution/grouping/copying/pasting/cutting/deleting, or use the shortcut keys.

| Name | Shortcut Key |
| --- | --- |
| Step-by-Step Execution | / |
| Group | ^ Ctrl + G |
| Copy | ^ Ctrl + C |
| Paste | ^ Ctrl + V |
| Cut | ^ Ctrl + X |
| Delete | ⌦ Del |

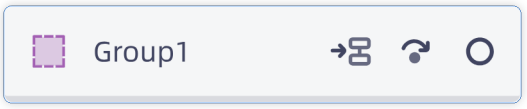
**Note**

Connections between commands will be copied along with the commands.

Select a command on the canvas and edit the command parameters in the parameter panel on the right.

## Group Command

To simplify complex workflows and achieve repetitive tasks, Robotscan Control allows users to group multiple commands.




Grouping operation: Select multiple commands on the canvas (supports multi-selection or box selection), and complete the grouping by choosing **Group** from the right-click menu or using the shortcut ^ Ctrl + G.

Loop Control: After grouping commands, you can set the number of times the commands within the group will be executed through the parameter panel on the right.

Renaming a group: After selecting the group, you can right-click or press the F2 shortcut key to rename it.

Edit Group: Click the edit group command icon to view and edit the commands within the group in a new view.

## Execute Command

1. Choose the execution method:
   - **Step-by-Step Execution**: Execute the currently selected command and stop. This mode is typically used for debugging purposes, allowing you to execute each command in the workflow step by step, checking the output at each step to identify errors or optimize the process.
   - **Continuous Execution**: Execute the first command and continue executing the next command until the last command is completed.
2. Set the number of executions (optional):

   If you choose the **Continuous Execution** method, you can check **Loop** and enter the number of loops as needed. After all commands are executed, it will continue to execute from the beginning until the loop count is completed.
3. Start Execution:
   - **Continuous Execution**: Click the ▶ **Start** button to execute the entire workflow.
   - **Step-by-Step Execution**: Click the ▶ **Start** button or double-click the command to execute the current step.
4. The timeline will display the current execution count and running time.

| Button | Description |
|---|---|
| ▶ Start | Start executing the current workflow. |
| ⏸ Pause | Pause the currently executing command. |
| ■ Stop | Stop the currently executing command and exit execution state. |

> ⚠ **Note**
>
> After starting execution, you cannot switch execution modes, add, connect, or edit commands.

## Debug Command

○ Skip: Temporarily ignore the execution of one or more commands during debugging, directly skipping these commands and continuing with subsequent commands.

Breakpoint: Set a breakpoint on a command during debugging; when execution reaches that command, the program will pause.

# Inspection

With workflow orchestration, the software automates the entire process of scanning, inspecting, and reporting. This enables fully automated, batch processing of inspections, quickly providing accurate 3D data of the workpiece.

After finishing the scan and processing and saving the scan data in a grid format, add the inspection command, such as SHINING 3D Inspect. Configure all the paths and check **PDF** and **Open Report (PDF)**. When the process reaches this command and executes successfully, it will automatically save and display the inspection report in the **Inspection View**.

## Workflow Example

## Inspection Report Example

# Command List

## PLC Commands

### Custom Signal On

Select Hexadecimal Address:

- 0x1008
- 0x1009
- 0x100A
- 0x100B
- 0x100C
- 0x100D
- 0x100E

### Custom Signal Off

Select Hexadecimal Address:

- 0x1008
- 0x1009
- 0x100A
- 0x100B

- 0x100C
- 0x100D
- 0x100E

#### Wait for Custom Signal

Select Hexadecimal Address:

- 0x1012
- 0x1013
- 0x1014
- 0x1015
- 0x1016
- 0x1017
- 0x1018
- 0x1019
- 0x101A
- 0x101B

Select Judgment Value: true/false

#### Custom Write

Select Hexadecimal Address:

- 0x3066
- 0x3067
- 0x3068
- 0x3069
- 0x306A
- 0x306B
- 0x306C
- 0x306D

## Turntable Command

#### Turntable Homing Routine

- Angle: 0 ~ 360°.
- Speed: 1 ~ 100%.
- Acceleration/ms/(60r/min): 80 ~ 9999.

#### Turntable Routine

- Angle: 0 ~ 360°.
- Relative Angle: Default is on, rotates the specified angle based on the current position.
- Speed: 1 ~ 100%.
- Acceleration/ms/(60r/min): 80 ~ 9999.

## Robot Commands

> **Note**
>
> - The point refers to the final coordinates where the robot stops moving.
> - When recording the point, ensure that the robot is in an enabled state.

#### Homing Routine

Get Current Position: Obtain the current position as origin.

#### Linear Motion

The robot's end effector moves along a straight line.

- Speed: 1 ~ 100%.
- Acceleration: 1 ~ 2500 mm/s².
- Linear Speed: 1 ~ 2000 mm/s.
- Transition Radius: 50 mm
- Teaching Interval: Points are recorded every 5 seconds by default, and can be set up to every 60 seconds.
- Robot Path Teaching:

  #### Manual

  After checking **Auto**, click the **Start Teaching**. Then physically move the robot arm to the desired position, and hold still for the duration set in the point recording interval; the software will add the point into point list.

#### 3Dconnexion SpaceMouse

When the 3D mouse is connected, it defaults to recording points with the 3D mouse.

Left-click to save the point, right-click to exit recording.

### Joint Motion

The robot's end effector moves along an irregular curve.

- Speed: 1 ~ 100%.
- Acceleration: 1 ~ 360°/s².
- Joint Speed: 1 ~ 180°/s, default 90°/s.
- Transition Radius: 50 mm
- Teaching Interval: Points are recorded every 5 seconds by default, and can be set up to every 60 seconds.
- Robot Path Teaching:

#### Manual

After checking **Auto**, click the **Start Teaching**. Then physically move the robot arm to the desired position, and hold still for the duration set in the point recording interval; the software will add the point into point list.

#### 3Dconnexion SpaceMouse

When the 3D mouse is connected, it defaults to recording points with the 3D mouse.

Left-click to save the point, right-click to exit recording.

### General Motion

Set the movement mode at each point. The default movement mode is joint motion, which can be modified to linear motion.

- Auto Scan: Once enabled, there's no need to add the **Start Scan** command; object will be scanned automatically upon reaching the designated position.
- Teaching Interval: Points are recorded every 5 seconds by default, and can be set up to every 60 seconds.
- Robot Path Teaching:

#### Manual

After checking **Auto**, click the **Start Teaching**. Then physically move the robot arm to the desired position, and hold still for the duration set in the point recording interval; the software will add the point into point list.

### 3Dconnexion SpaceMouse

When the 3D mouse is connected, it defaults to recording points with the 3D mouse.

Left-click to save the point, right-click to exit recording.

## Calibration Commands

### Enter Calibration

The first command of calibration.

### Start Calibration

Start calibration.

### Calibration Calculation

Calculate calibration.

### Quit Calibration

Quit the calibration.

### Deploy Calibration Board

Pop up the calibration board inside the base.

### Stow Calibration Board

Stow the popped-up calibration board back into the base.

## Scanner Commands

### Quick Scan Template

### New Scan Project

- Scan Mode: The scan mode is read-only and set to laser mode.
- Name: Project group name.

- [ ] Keep Backup:
  - If checked, a backup file will be generated each time you save.
  - If not checked, executing the command multiple times will only keep the last created project.

- Default Path: Set the default path for saving the project group.

#### Scan Setting

- Scan Mode:
  - Scan Mesh: This mode is used for directly scanning to generate mesh data, suitable for most scanning scenarios.
  - Global Markers: This mode is used to scan global markers data on the object's surface. By collecting markers, you can quickly obtain the global markers data of the object. After scanning, you can save the data or switch to scan mesh mode to continue scanning for added convenience.
- Resolution: The smaller the point distance, the more details in the model; conversely, the larger the point distance, the fewer details in the model.
- Light Source Mode:
  - 50 Lines: Mainly used for quick scanning of large objects.
  - 7 Lines: Mainly used for fine scanning.
  - 1 Line: Mainly used for deep hole and gap scanning.
- Object: Choose different scanning object modes based on the material of the object, including **Normal** and **Reflective**.
- Brightness: Drag the slider to adjust the scanning brightness until a more complete laser data is reconstructed. If the brightness is too high, there will be more noise in the scanning data.
- Outdoor Mode: When enabled, it allows normal scanning to obtain object data in strong light environments (such as outdoors).

> **Note**
>
> Please avoid direct sunlight when scanning objects.

- Data Quality Indicator: Displays scanning quality in the form of a spectrum, where blue represents high-quality scanned data, and yellow indicates insufficient scanning that requires further scanning. Insufficiently scanned data may disappear or show abnormalities after data processing.
- Scan in Point Cloud: When enabled, the scanning process will display point cloud data, reducing memory consumption.
- Local Enlarged View: When enabled, the scanning interface will only display a local view of the scanned object, which can be used for supplementary scanning of small holes.

- View Lock: When enabled, the 3D scene in the scanning interface will no longer change with the movement of the scanner, which can be used to scan the current face of the object.

#### Import Global Markers File

Select the local path for the global markers file (.p3, .txt, .asc, .dgm).

#### Global Markers Optimization

Optimize the global markers data.

#### Start Scan

Start scanning.

#### Pause Scan

Pause scan.

#### Clear Data

Delete all scan data in the current project.

#### Mesh Optimization / Mesh Processing

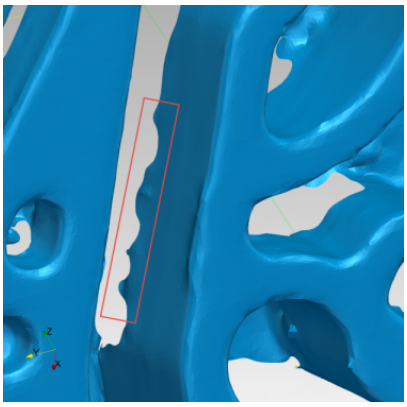
Packaging Mode: Choose **Mesh Processing** or **Mesh Optimization**.

Mesh Parameters:

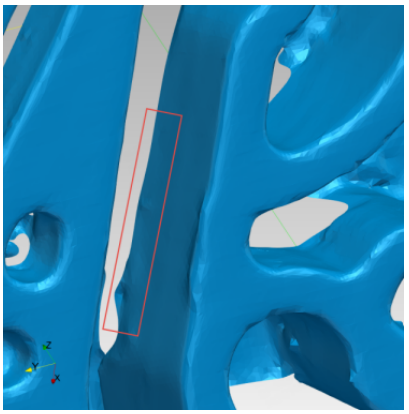
| Icon | Function | Description |
|---|---|---|
| | Unwatertight | Unclosed model stays the way it is scanned, and its processing time is shorter than that of **Watertight**. |
| | Half watertight | Some of the holes will be filled automatically. |
| | Watertight | All holes will be filled automatically. The data can directly be 3D printed.<br>The **Model Quality** can be set as **High**, **Med** (default), or **Low** only for the watertight mesh. |

**Mesh Processing:**

- **Optimization**: Optimizes data and reduces noise. Higher settings may lose some small details.
  - None: No optimization.
  - Standard: Light optimization, trying to retain surface features of the data.
  - Medium: Reduces noise on the data surface.
  - High: Reduces noise while making the data smoother.
- **Smooth**: Smooth the possible noise on the surface of the scan data. Three optimization options are available: **Standard** (default), **Med** and **High**.
- **Remove Isolated Surface**: Remove small floating parts isolated from the main data. Adjust the area of floating parts to be removed by dragging the slider or using the stepper. The default value is 10, and the range is 0 ~ 1000 (0 indicates no removal). **For a specific illustration of the effect, refer to the following images:**
- **Edge Optimization**: By checking it, the boundary data will be more complete and smoother after meshing.



Before



After

- **Maximum Number of Patches**: Set the max plate number to get mesh model's triangle plate number which is within configured plate number (enabled by default). Please input the value reasonably, avoiding entering too small values, as excessive simplification may result in lower data quality.
- **Automatic Hole Filling**: Auto fill the small holes when generating the mesh. The default value is 10 with a range of 0 to 100. You can delete unwanted neighboring areas to delete edge noise data; the default value is 3 with a range of 0 ~ 10 (0 indicates no neighboring area deleted). This function is unavailable for watertight models.
- **Remove Spike**: Remove spike-like data on the image edge.
- **Marker Hole Filling**: Fill in the surface of the object that is not scanned to the pasting markers. For the align mode of **Markers**, this function is enabled by default for watertight models; for other align modes, this function is unavailable.
- **Recommended Parameters**: When enabled, it will automatically use the recommended parameters for meshing.

#### Save Data

Save Types: Supports STL, P3, ASC, OBJ, PLY, 3MF, and DGM formats.

- Name: The name of the model file.
- [ ] Keep Backup:
  - If checked, a backup file will be generated each time you save.
  - If not checked, multiple executions of the command will only keep the last created project.
- [ ] Use SN:

  After connecting to the external controller, the product SN will be transmitted to the RC software via the Modbus protocol, and the SN will be appended to the custom name when saving the STL file.
- Save Path: Click **Select** to choose a folder in the file explorer as the save path for the model file.

## Inspect Commands

#### SHINING3D Inspect

- exe Program Path: Select the path where the SHINING3D Inspect exe program is located.
- Workspace path: Select the project file for SHINING3D Inspect (*.SNIProj), that is, inspection template.
- Report Export Path: Select the path to save the inspection report.
- Report Export Format
  - [x] PDF
    - [x] Open Report (PDF): Automatically open the inspection report in the Inspection interface after scanning is complete.
  - [ ] CSV
- [ ] Save inspection project
- [ ] Share Memory

#### PolyWorks

- Inspection Software: The default inspection software is PolyWorks.
- exe Program Path: Select the path where the PolyWorks exe program is located.
- Workspace Path: Select the path for the PolyWorks workspace.
- Report Export Path: Select the path to save the inspection report.
- Macro script: Select the path where the PolyWorks macro script is located.
- [x] Open inspection report

#### MES System Interface

Call external programs with parameters.

- External exe path: Select the path where the MES client exe file is located.
- External exe custom parameters: Add startup parameters. Special characters are not supported.

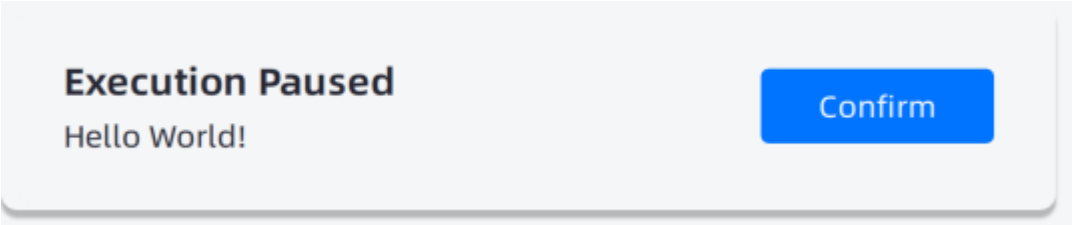
## Other Commands

#### Waiting Time

Insert the specified wait time (ms) between commands.

#### Custom Prompt

Input text of 1 to 30 characters. When this command is executed, a text prompt will pop up, and the execution process will pause. Click confirm to continue the execution process.

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